

SANKUR KUNDU

Software Engineer · Machine Learning Engineer

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Engineer who ships systems end to end. Co-founder and Head of Engineering at Gigzen: built [three Android apps](#), [Waggle Gig](#), [Waggle Business](#) and [Waggle](#), for riders, shops and customers on one Postgres backend, and Populace, a multi-user testing platform on npm.

EXPERIENCE

Gigzen · Co-founder & Head of Engineering

Bhubaneswar | Apr 2026 – Present

- Co-founded Gigzen (incorporation in progress); lead all engineering, schema to signed release: one TypeScript codebase building three Android apps on 53 Postgres tables with **row-level security on every one** (90 policies, 154 functions).
- [Waggle Gig](#) (riders): real-time job offers with a 45-second window, dispatch decided first-accept-wins inside the database, voice turn-by-turn with rerouting and a snap-to-road algorithm, and earnings per hour.
- [Waggle Business](#) (shops): orders turned into rider jobs with 4-digit handover codes, a live map of its riders, numbered GST invoices, UPI rider settlement by UTR, and month-end billing on pg_cron.
- [Waggle](#) (customers): nearby shops, menus with item choices, coupons, scheduled and group orders, and live tracking with the rider snapped to the road and an ETA counted down from the distance left.
- Tested it as one product: a Playwright pipeline drives one order through all three apps' screens in 18 steps, and 134 attacks on the database rules are all blocked. Debugged a render loop firing ~450 API calls a second.
- [Populace](#): built a multi-user testing platform that drives an app through its own API as simulated, authenticated users. Its first run found **5 defects in 3.5 minutes**, 3 of them in an Android app that had passed full manual testing.
- App-agnostic engine behind a 13-method adapter contract, proven on Gitea, with per-endpoint p50/p95 latency and three guards against running on production; [121 self-tests](#), zero runtime dependencies. Ships on [npm](#) and as an Electron app.
- Load-tested with **200 concurrent simulated drivers: 932,455 API calls, zero failures, p95 under 900 ms**; across 3,174,576 calls in one day, measured scalability from 178 to 272 calls/sec at 100 to 300 drivers.

Freelance Data Scientist · Contract via Upwork

Remote, client in Spain | Jun – Jul 2026

- Merged two Spanish hotel censuses (5,909 and 8,058 records; 118 and 281 columns) on national statistics identifiers in Python and Pandas, over Excel and STATA .dta sources.
- Built a Google Trends panel of 10,038 city-months: deseasonalised, z-scored and flagged for positive shocks.
- Collaborated cross-functionally with researchers and statisticians on validation and reproducibility checks.

SELECTED ML PROJECTS

- **ML Serving Platform** — A/B routing, circuit breaking and drift detection around inference; Triton, MLflow, Kubernetes.
- **Edge Model Compression** — distillation, pruning, int8 ONNX; targeted 8× smaller, measured -6.3%, published why.
- **IoT Predictive Maintenance on GCP** — LSTM failure prediction on a leak-free chronological split; BigQuery, Vertex AI.

SKILLS

Languages: Python, TypeScript, Java, SQL, R, STATA

ML & AI: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy; deep learning (LSTM, BERT, DDPM), training and evaluation pipelines, statistical modeling, distillation, pruning, int8 quantization, ONNX, anomaly detection, LLM-assisted diagnosis

Systems & Cloud: System design, distributed systems, CI/CD, GCP (BigQuery, Vertex AI, Dataflow, Pub/Sub), AWS (Lambda, S3), FastAPI, Flask, Node.js, React, Capacitor, Electron, Postgres, Supabase, Redis, Docker, Kubernetes, Terraform, Playwright

EDUCATION & CERTIFICATIONS

Institute of Engineering and Management, Kolkata · B.Tech, Computer Science & Engineering

2022 – 2026

Certifications: Meta Data Analyst; Google Data Analytics; Google Cybersecurity; IBM Cybersecurity Essentials; Google Cloud and AWS engineering paths; Microsoft Azure AI Cloud Apps